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Death by Horse Kick: A Bayesian Tale from the Prussian Army

What a 19th-century military dataset can teach us about probability, belief, and uncertainty



Rob Taylor, PhD Follow 10 min read · Jan 10, 2023

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Photo by Pawel Czerwinski on Unsplash

Updated April 2026 — this post has been substantially revised and expanded from its original version.

Introduction

Imagine getting kicked by a horse. That's got to hurt, right? Any way you look at it, it's going to be a bad day. So perhaps we should spare a thought for the soldiers in the Prussian army actually *killed* by a horse kick — I mean, what a way to go.

But here's what makes those deaths more than a historical curiosity: they follow a pattern. A suspiciously neat, mathematically elegant pattern. And the man who first noticed this — a Russian-born statistician named Ladislaus von Bortkiewicz— used it to make a case that would quietly reshape how we think about rare events and probability.

Von Bortkiewicz was, by conviction, a frequentist. He believed that probability was a property of long-run frequencies, not of degrees of belief. And yet the dataset he published in his 1898 book *The Law of Small Numbers* became one of the most celebrated demonstrations in all of applied statistics[1] — and one that Bayesians would later find rather congenial. There's a pleasing irony in that.

The data was collected over a 20-year period (1875–1894) and recorded the number of officers killed each year by horse kicks across 14 cavalry corps within the Prussian army. In total, 196 deaths were documented from 280 observations (14 corps × 20 years), giving an average rate of 196/280 = 0.70 deaths per corps per year. Less than one death per year, per corps — a rare event, by any measure.

What is quite remarkable is that this data is approximated very well by a Poisson distribution. The Poisson typically models the number of events that occur within a fixed interval of time or space, and it is especially well-suited to rare events occurring within large populations or systems. That the messy, contingent reality of horse kicks in the Prussian military should conform so neatly to a mathematical distribution is, on reflection, a striking thing. It suggests that randomness itself has a shape.

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

The Poisson probability mass function (PMF) (image by author).

Von Bortkiewicz used the maximum likelihood estimate for the Poisson rate parameter λ to demonstrate the correspondence between the expected frequencies and the actual deaths. But that's not what we're going to do here. Instead, we're going to take a Bayesian approach — one that goes beyond fitting a single number and asks instead: *what should we believe about λ, given the data we've seen?*

Building a Bayesian Model from First Principles

In a Bayesian analysis, parameters are not treated as fixed, unknown constants to be estimated — they are treated as random variables, carrying uncertainty that we can quantify and update. The core idea is that we begin with a *prior distribution* that encodes our beliefs about a parameter before seeing any data. We then combine that prior with the *likelihood function* — which captures how probable the observed data is under different parameter values — to produce a *posterior distribution* that reflects our updated beliefs having seen the data.

Using Bayes' rule, the posterior is:

$$\pi(\Theta|D) = \frac{\pi(D|\Theta)\pi(\Theta)}{\pi(D)}$$

Bayes' rule (image by author).

where D is the observed data and Θ is the parameter to be estimated. The denominator — the marginal likelihood — is just a normalising constant ensuring the posterior integrates to one. Because it doesn't involve the parameter of interest, it is often dropped, leaving the proportional form that is the workhorse of Bayesian computation:

$$\pi(\Theta|D) \propto \pi(D|\Theta)\pi(\Theta)$$

The unnormalized posterior is proportional to the likelihood times the prior (image by author).

The Likelihood

Let $\mathbf{Y} = (k_1, k_2, k_3, \ldots, k_n)$ denote the set of $n = 280$ observations. Assuming each observation is independent — that is, events occur randomly across both years and cavalry corps, with each soldier equally susceptible — the likelihood function is:

$$\pi(\mathbf{Y}|\lambda) = \prod_{i=1}^{n} p(k_i|\lambda) = \prod_{i=1}^{n} \frac{\lambda_i^k e^{-\lambda}}{k_i!}$$

The likelihood function for the Gamma-Poisson model (image by author).

The denominator contains a term that doesn't involve λ, so we can drop it and work with the unnormalized likelihood:

$$\pi(\mathbf{Y}|\lambda) \propto \lambda^{\sum_{i=1}^{n} k_i} e^{-n\lambda} = \lambda^{n\bar{k}} e^{-n\lambda}$$

The unnormalized likelihood, having dropped the constant term in the denominator (image by author).

where:

$$\bar{k} = \frac{1}{n} \sum_{i=1}^{n} k_i$$

Arithmetic average of the data (image by author).

Choosing a Prior

Here is where Bayesian analysis asks something that frequentist methods do not: *what do we believe about λ before looking at the data?*

For the horse kick problem, this is actually a question worth sitting with. We know from context that deaths are rare — the Prussian military was not in the habit of losing officers to their own horses at high rates. A prior that places most of its mass at low values of λ seems sensible. We also want a prior that is *conjugate* to the Poisson likelihood, meaning that the posterior will belong to the same distributional family as the prior — a mathematical convenience that allows us to derive the posterior in closed form, without needing numerical methods.

For the Poisson model, the conjugate prior is the Gamma distribution, with shape parameter α and rate parameter β:

$$\pi(\lambda) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$$

Probability density function (PDF) for the Gamma distribution (image by author).

Dropping the normalising constant, this becomes:

$$\pi(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda}$$

Unnormalized Gamma distribution (image by author).

Notice that the prior and the likelihood share the same functional form — both involve a power of λ multiplied by a negative exponential. This is

precisely why conjugacy works: the two expressions multiply cleanly.

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For our prior, we'll set α = 1 and β = 1, giving a Gamma(1, 1) distribution — which is equivalent to an exponential distribution with rate 1. This is a weakly informative prior: it places mass across a broad range of λ values, with a gentle preference for smaller ones. It reflects the sensible belief that deaths per corps per year are more likely to be near zero than near ten, without being so dogmatic that the prior overwhelms the data. It is, in other words, an honest starting point.

Deriving the Posterior

Multiplying the likelihood by the prior and collecting like terms:

$$\begin{aligned}\pi(\lambda|\mathbf{Y}) &\propto \pi(\mathbf{Y}|\lambda)\pi(\lambda) \\ &\propto \lambda^{n\bar{k}}e^{-n\lambda}\lambda^{\alpha-1}e^{-\beta\lambda} \\ &\propto \lambda^{n\bar{k}}\lambda^{\alpha-1}e^{-n\lambda}e^{-\beta\lambda} \\ &\propto \lambda^{\alpha+n\bar{k}-1}e^{-(\beta+n)\lambda}\end{aligned}$$

Deriving the unnormalized posterior distribution by collecting terms (image by author).

This expression has the kernel of a Gamma distribution. Because the Gamma is the conjugate prior for the Poisson, the normalised posterior is also a Gamma — specifically:

$$\pi(\lambda|\mathbf{Y}) \sim \text{Gamma}(\alpha + n\bar{k}, \beta + n)$$

The posterior distribution with updated shape and rate parameters (image by author).

That's a strikingly clean result. All the data contributes through just two sufficient statistics: the total number of deaths (which updates the shape parameter) and the total number of observations (which updates the rate parameter). The posterior is fully determined by what we knew before and what we've now seen.

Fitting the Horse Kick Data

Plugging in α = 1, β = 1, $n = 280$, and the total number of deaths $n\bar{k} = 196$:

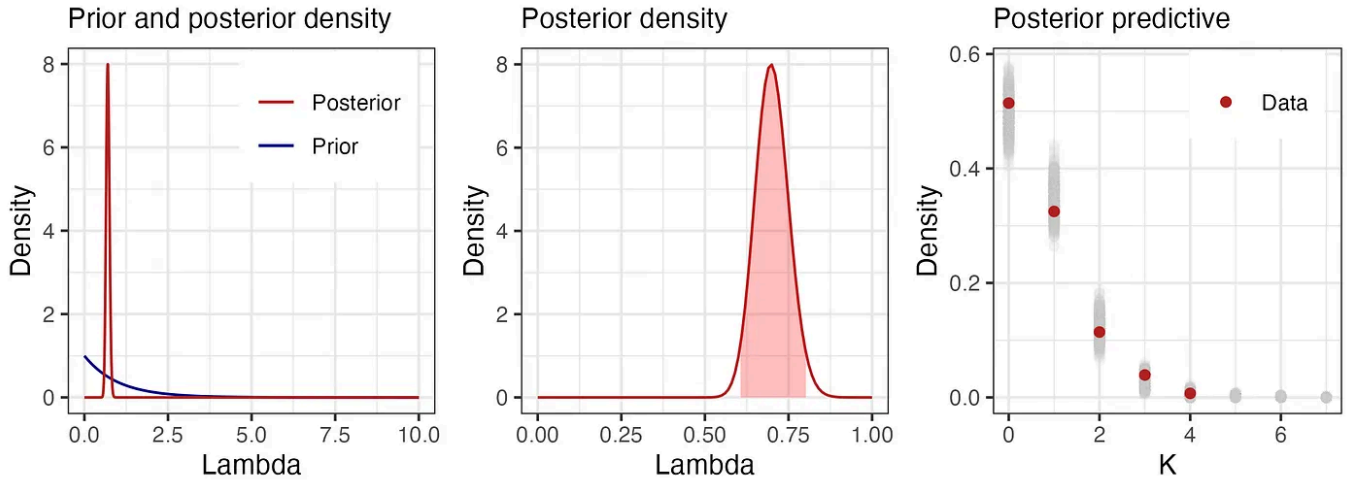
$$\begin{aligned}\pi(\lambda|\mathbf{Y}) &\sim \text{Gamma}(\alpha + n\bar{k}, \beta + n) \\ &\sim \text{Gamma}(1 + 196, 1 + 280) \\ &\sim \text{Gamma}(197, 281)\end{aligned}$$

Gamma posterior using the horse kick data (image by author).

And that's it!

The posterior has a mean of 197/281 ≈ 0.701 — almost exactly the maximum likelihood estimate of 196/280 = 0.7. This is reassuring but also expected: with 280 observations, the data speaks loudly, and a weakly informative prior has little influence on the outcome. Had we started with a very different prior — say, one centred at λ = 5 — the posterior would have been pulled in that direction. The fact that it wasn't is the data asserting itself.

The posterior is plotted below alongside the prior (left panel). The contrast is instructive: the prior is broad and diffuse, spreading mass across a wide range of λ values. The posterior, having absorbed 280 observations, is narrow and sharply peaked. Our uncertainty about λ has been dramatically reduced.



Left panel: Prior and posterior densities; Middle panel: Posterior density with 95% Credible Interval shaded; Right panel: Posterior predictive distribution and corresponding data (image by author).

The shaded region in the middle panel is the 95% *credible interval*, running from approximately 0.606 to 0.802. This is not the same as a frequentist confidence interval, though the two can look numerically similar. The distinction is conceptual but important: because the posterior is a probability distribution, we can make direct probability statements about λ. The credible interval tells us that we should assign 95% probability to λ lying in that range. A confidence interval, by contrast, is a procedure — one that, in repeated sampling, will contain the true (fixed) parameter 95% of the time. It says nothing about the probability that any particular interval contains the true value.

Assessing Model Fit

The final piece of any Bayesian analysis is checking whether the model actually fits the data. To do this, we derive the *posterior predictive distribution* — the distribution of new observations, averaged over our posterior uncertainty in λ.

The idea is straightforward: for each plausible value of λ in the posterior, compute the Poisson probabilities of new data under that rate, then average across all those values. This averaging integrates out our uncertainty about λ, producing a marginal distribution over new observations that honestly accounts for what we don't know.

For the Gamma-Poisson model, this integration has a closed form. With updated posterior parameters $\alpha' = \alpha + n\bar{k}$ and $\beta' = \beta + n$, the posterior predictive is a Negative Binomial distribution:

$$\pi(\tilde{Y}|\mathbf{Y}) \sim \text{NegBinom}\left(\alpha', \frac{\beta'}{\beta' + 1}\right)$$

The posterior predictive distribution for the Gamma-Poisson model is the Negative Binomial distribution (image by author).

The right panel of the figure above plots this distribution against the observed data. Rather than just overlaying the distribution, I simulated a large number of datasets from the posterior predictive and compared them to the actual observations. The fit is good — the model captures the shape of the data well, with most of the mass concentrated at zero and one death per year, and a long, thin tail for rarer events.

Von Bortkiewicz would have recognised the fit. He just would have arrived at it differently.

A Note on Priors

A natural question at this point: what if we had chosen different prior parameters?

It depends on what those values are. A broad, flat prior — like the one we used — will be quickly overwhelmed by data, and the posterior will closely track the likelihood. A narrow, peaked prior reflects strong prior conviction and will resist updating unless the data is very inconsistent with it. When prior and data conflict, the posterior splits the difference, weighting each source of information by its relative precision.

This is the part of Bayesian analysis that makes some people uncomfortable. Choosing a prior feels subjective — and it is, to a degree. But subjectivity in

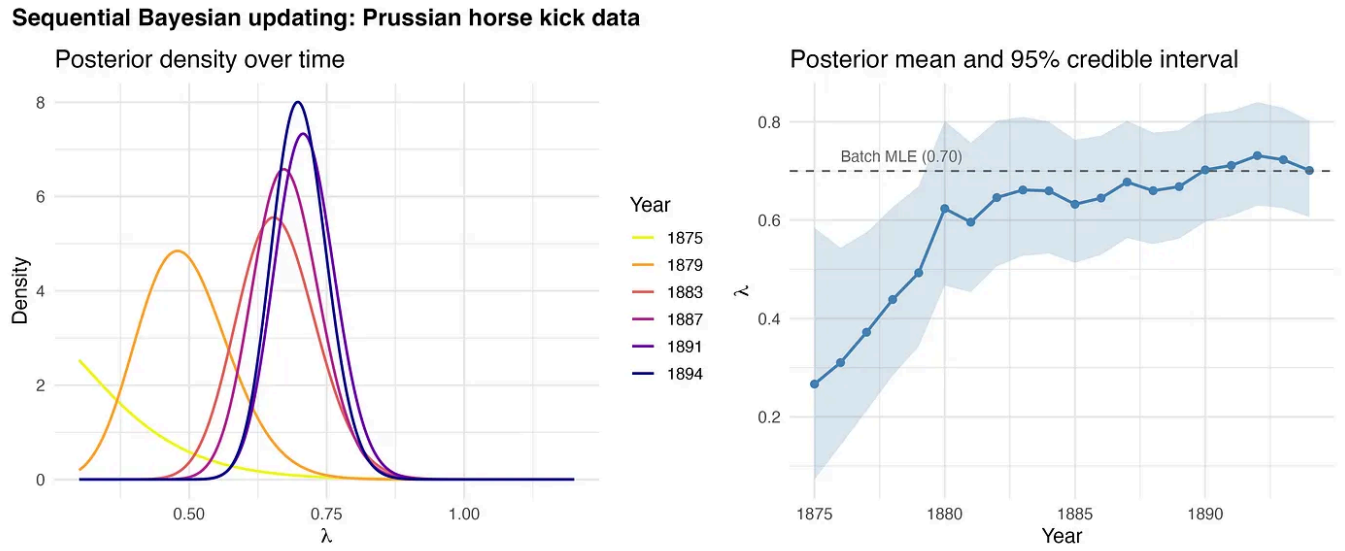
this context doesn't mean arbitrariness. A well-chosen prior reflects genuine background knowledge, and the choice should be reported, justified, and subjected to sensitivity analysis: how much does the posterior change if you vary the prior? For well-informed models with substantial data, the answer is usually "not much." For sparse data or high-stakes inference, it can matter considerably.

There is an old adage in Bayesian circles: *yesterday's posterior is today's prior*[2]. As new data arrives, what we've already learned becomes the starting point for the next analysis. It is a coherent, cumulative approach to knowledge — and in that sense, rather more honest about how scientific understanding actually develops than a framework that pretends to begin each analysis from scratch.

Updating Beliefs Year by Year

So far we've fed all 280 observations into the model at once and read off the posterior. But the data wasn't collected that way — it arrived year by year, across two decades. And one of the more satisfying features of the Bayesian framework is that we can honour that structure: start with our Gamma(1, 1) prior in 1875, update on that year's deaths, carry the resulting posterior forward as the prior for 1876, and so on. The "yesterday's posterior is today's prior" principle, made literal.

The update rule is the same each year. If we enter year t with a Gamma(α, β) prior and observe a total of k deaths across the 14 corps, we exit with a Gamma($\alpha + k$, $\beta + 14$) posterior — the shape absorbs the deaths, the rate absorbs the number of observations. Nothing more complicated than that.



Left panel: The change in the posterior density over time, thinned to every four years. Right panel: the change in posterior mean with 95% credible intervals (image by author).

The left panel below shows the posterior density evolving across all 20 years. In 1875 the distribution is broad, reflecting genuine uncertainty about λ. As each year's data arrives, the posterior tightens — the distribution narrows, its peak sharpens, and the influence of any single year's observation diminishes. By the mid-1880s the posterior has largely settled, and the final years contribute only minor refinements.

The right panel tracks the posterior mean and 95% credible interval over time. Early on, the estimate moves around noticeably as each new year carries real weight. By the end, the credible interval has narrowed considerably and the mean has converged to approximately 0.70 — consistent with what we found fitting all the data at once, as it should be. Sequential updating and batch updating are mathematically equivalent; what the sequential view adds is intuition about *when* the data has said most of what it has to say.

Thanks for reading. If you want to play around with the prior specification and see how it affects the posterior, the code for this analysis is available on GitHub. All it

takes is changing two numbers.

A note on this revision: *the original version of this post focused primarily on the mechanics of the Gamma-Poisson model. This updated version expands the historical framing around von Bortkiewicz and adds a section on sequential Bayesian updating to illustrate how beliefs evolve as data accumulates year by year.*

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References

[1] Ladislaus von Bortkiewicz, *Das Gesetz der kleinen Zahlen* [The law of small numbers] (Leipzig, Germany: B.G. Teubner, 1898). This book also contained

another rather grim analysis of the rate of child suicide in Prussia. I think we'll leave that one.

[2] Lindley, D. V. (2000). The philosophy of statistics. *The Statistician*, *49*, 293–337.

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What are your thoughts?



Matthias Wiedemann

Feb 8, 2023

Nice! Horse kick is great!

1 Reply



Michaelaoash
Jul 29

This is a great and helpful essay for thinking both about Poisson distributions and Bayesian statistics.

Two brief thoughts.

1) There may be a typo in one of the equations. In the first equation of the section "The Likelihood" the subscript *i* should be on the *k* not on the *lambda* in the last expression. My attempt at corrected LaTeX is below:

\pi(\mathbf Y|\lambda) = \prod_{i=1}^{n} p(k_i|\lambda)= \prod_{i=1}^{n} \frac{\lambda ^ {k_i} e^{-\lambda}} {k_i!}

2) You wrote this so elegantly that it made me think better about the prior. It might be helpful to note:

- the Gamma distribution has non-negative support, i.e., (0, \infty) and show a picture earlier rather than later.

- the expected value of a Gamma(alpha,beta) distribution, which corresponds to the expected MLE of lambda in the Poisson being modeled, is alpha / beta.

- that there are many values of alpha and beta with the same ratio alpha / beta, and the difference is the concentration or diffuseness of the prior. In the example you gave alpha = 1, beta = 1 as the prior (which corresponds to alpha/beta = 1 = expected MLE of lambda with relatively diffuse priors. But alpha = 10, beta = 10 has the same MLE (alpha / beta = 1) but represents a lot more certainty about the expectation (and all the more so for alpha = 100, beta = 100). The data will move the posterior much less (or much slower in relation to the availability of new data) with higher values of priors on alpha and beta (with the same ratio of the priors on alpha and beta).

I'm sorry if my note is not clear. Your write-up helped me a lot. Thank you very much.

Reply



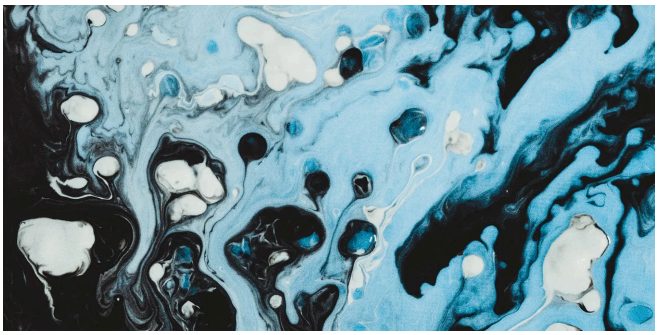
Michael Friendly
Mar 6, 2023

> Fitting the Horse Kick Data

It would be helpful if you posted the code (or a link to) for this example.

1 reply Reply

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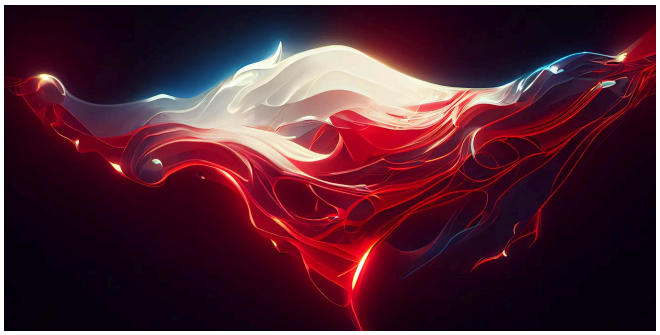


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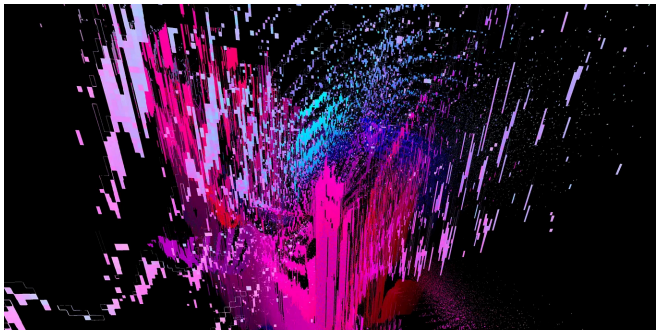


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